

**Synthesis of a Coordination Compound Containing Iron
and
Analysis of this Compound Using Redox Methods
Prelab**

Name _____

Total _____/10

NOTE: AT THIS POINT YOU WILL ANSWER ALL PRELAB QUESTIONS IN YOUR CARBON COPY LAB NOTEBOOK. BE SURE TO INCLUDE THE NAME OF THE EXPERIMENT AND THE QUESTIONS ASKED. AT THE START OF LAB YOU WILL HAND IN THE CARBON COPY, WHICH WILL CONTAIN ANSWERS TO ALL QUESTIONS.

1. What is the purpose of this experiment?
2. What colour should the crystals you make be?
3. Write the generic chemical formula for the coordination compound you will be making in this laboratory experiment.
4. What are the names of the ions that are in your coordination compound?
5. What are the charges of each ion in the complex that you are going to make in this experiment?

Synthesis of a Coordination Compound Containing Iron and Analysis of this Compound Using Redox Methods

You will synthesize a complex salt containing iron and then analyze it using redox volumetric methods to determine the percent composition and ultimately, its empirical formula. There will be two parts to this laboratory experiment: Part I: the synthesis of the complex salt and Part II: the analysis of the complex salt. The laboratory experiment will be performed over a three week period.

Introduction

Ions of the transition elements exist in aqueous solution as complex ions. A complex ion is a metal ion with a molecule called a ligand attached to it through coordinate covalent bonds. A complex salt is an ionic compound but it differs in the fact that there are these covalent bonds attaching the metal to the ligand. The iron(II) ion, for example, exists in water as $\text{Fe}(\text{H}_2\text{O})_6^{2+}$. The water molecules in this ion are arranged about the iron atom with their oxygen atoms bonded to the metal by donating electron pairs to it. Replacing the H_2O molecules with six CN^- ions gives the $\text{Fe}(\text{CN})_6^{4-}$ ion. A coordination compound is a compound consisting either of complex ions and other ions of opposite charge or a neutral species. An example of a coordination compound is the compound $\text{K}_4[\text{Fe}(\text{CN})_6]$ which contains the complex ion $\text{Fe}(\text{CN})_6^{4-}$ and four K^+ ions. Some of the transition elements have biological activity and their role in human nutrition depends in most cases on the formation of complexes and coordination compounds.

In the first part of this experiment, you will make a complex salt containing iron in which water molecules and oxalate ions ($\text{C}_2\text{O}_4^{2-}$) act as ligands. The formula for this complex will have the formula $\text{K}_x\text{Fe}_y(\text{C}_2\text{O}_4)_z \cdot 3\text{H}_2\text{O}$. After this salt has been prepared you will determine its percent composition in iron and oxalate ion and from this information, determine the values for x, y and z in the formula written above.

In the synthesis of the complex salt, you will dissolve an iron(III) compound and form the coordination compound. When your coordination compound is crystallized, you can begin the analysis of this compound. The overall reaction for this synthesis is:



It is assumed that $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ is the limiting reagent and when the number of moles of iron (y) are determined from the percent composition of the coordination compound above, the equation can be balanced and the theoretical yield of this compound can be determined.

Part I

Synthesis of the Complex Salt containing Iron

Procedure

YOU MUST COMPLETE THIS SYNTHESIS IN THE FIRST LABORATORY PERIOD SO THE COMPOUND CAN CRYSTALLIZE BETWEEN LABORATORY MEETINGS.

1. In a 250.0 mL beaker, weigh out approximately 10.00 g of potassium oxalate monohydrate, $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ on the **top loader** balance and add 20.00 mL of distilled water.
2. Heat this solution, while stirring, until all of the potassium oxalate has dissolved.
3. While the potassium oxalate solution is heating, you can make your solution of iron (III) chloride hexahydrate. In a 250.0 mL beaker, weigh out approximately 5.30 g of iron(III) chloride hexahydrate, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$. Be sure to use the **top loader** balance and record the mass to the nearest or 0.01g.
4. Dissolve the iron(III) chloride hexahydrate in 15.00 mL of water.
5. Filter the iron(III) chloride hexahydrate solution from step 4 into the hot potassium oxalate solution.
6. Place the beaker in your desk and cover it with an inverted beaker that's larger in size and it will stay in your drawer until the next laboratory period.

When you look at your beaker the next laboratory period, you should find green crystals covering the bottom and sides of the beaker. You will treat these crystals in the following manner.

7. Prepare a suction filtration apparatus as shown by your instructor and be sure that the filter flask is very clean. You will be using the liquid, called the filtrate, that pours into it to wash your crystals.
8. Use a rubber policeman to scrape the crystals and solution into the funnel. If you have any crystals that remain in the beaker, wash them into the funnel with some of the filtrate from the filter flask.
9. Rinse the crystals the crystals while they are in the funnel with 15 mL of a 1:1 alcohol-water solution. Pour this solution slowly so all of the crystals are washed.
10. Pour three 10 mL portions of acetone over the crystals so they will dry rapidly. It should not take more than 15 or 20 minutes for the crystals to thoroughly dry. **BE CAREFUL WITH**

THE ACETONE SINCE IT IS HIGHLY VOLATILE AND FLAMMABLE. DO NOT USE ACETONE IF THERE IS AN OPEN FLAME BY YOUR WORK AREA!!!!

11. Using the top loader balance, zero a clean, dry beaker. Place all of your dry crystals in the beaker and record the mass of the crystals you produced. You will then transfer your crystals to a clean dry weighing vial. Wrap your vial in aluminum foil since the crystals will decompose with the presence of light.

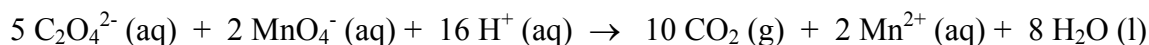
Part II

Analyzing the Complex Salt

There are three sections to part II of this experiment. Part A involves standardizing your potassium permanganate solution which you will use in the next two parts of the experiment. Part B involves determining the amount of oxalate and in part C involves analyzing the amount of iron in your coordination compound.

Part A: Standardizing a Potassium Permanganate Solution

In order to determine the amount of iron and oxalate in your compound, you must know the concentration of the potassium permanganate (KMnO_4) you are going to use in parts B and C. You will do this by titrating your permanganate solution into a solution containing a known mass of sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$). The titration reaction is:



If you need a review of titration procedures, refer back to the laboratory experiment Exploring Acid-Base Reactions.

Procedure

It is very important that all of your apparatus be clean and that distilled water is used in all solutions.

For your titrations (in parts A, B and C), always have a wash bottle with distilled water in it ready to use.

1. Collect 50.0 mL of the potassium permanganate solution that you are going to standardize.
2. Using the **analytical** balance, weigh approximately 0.1200 g of sodium oxalate and place in a 250 mL Erlenmeyer flask. Add 10 mL of 6 M H_2SO_4 to the flask and this will help the sample to dissolve. Add 65 mL of water to the Erlenmeyer flask.
3. Clean and rinse your buret. When the proper procedure for titration preparation has been performed (if unsure refer to the laboratory experiment Exploring Acid-Base Reactions) fill

the buret to the top with the KMnO_4 solution. Note that the solution is a very dark purple colour so you should take your volume readings from the top edge of the liquid instead of the bottom of the meniscus.

4. Heat the solution of sodium oxalate in the Erlenmeyer flask to 80-90 °C. When you remove the thermometer to perform the titration, be sure to rinse the thermometer into the flask since you do not want to lose any of the sodium oxalate.
5. Record the initial reading on the buret, to the nearest 0.01 mL and begin to add the KMnO_4 solution to the flask but do not add too rapidly and be sure to swirl the solution. You should observe that the purple solution loses its colour as it falls into the hot solution.
6. If you add the KMnO_4 solution too rapidly, or do not swirl well, you may find you have some brown colouration in your solution. This is due to the formation of manganese dioxide (MnO_2). If you have not added any more KMnO_4 than needed to reach the endpoint, the excess oxalate should reduce the MnO_2 momentarily. However, if you fail to swirl the sample and overshoot the endpoint while MnO_2 is formed, the titration is ruined and must be performed again.
7. You should start to notice that as you are nearing the endpoint of your titration that the decolouration of the KMnO_4 takes longer and longer. At this time you should add the KMnO_4 more slowly, preferably drop by drop. When you have reached the endpoint, there will be a faint colour that persists in the solution. If you are afraid that you will overshoot the endpoint with one more drop, then record the volume before you add that one drop and that way if you do overshoot, at least you have the final volume recorded and you will not have to perform an additional trial of the titration.
8. Repeat this procedure two more times.
9. It is useful to run what is called a blank for this titration since the sulphuric acid solution may contain some impurities that would react with the potassium permanganate and bring error to your data. Add 10 mL of 6 M H_2SO_4 to the flask and then add 65 mL of water to the Erlenmeyer flask and heat it to 80-90 °C. This should not take more than one or two drops so you may use your final volume from your last titration as the initial reading for this titration. Titrate until you have a persistent faint pink colouration.
10. Subtract this volume from the volume of KMnO_4 used in the titration of the sodium oxalate sample. Use this corrected volume for all of your calculations in part A.

Part B: Determination of the amount of Oxalate in the Complex Salt

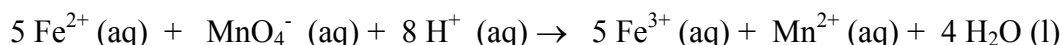
The determination of oxalate in the complex salt involves oxidation-reduction or redox titrations. The KMnO_4 will oxidize all of the oxalate to CO_2 which escapes. You will use the same chemical equation from part A to calculate the amount of oxalate in the complex salt.

Procedure

1. Using the **analytical** balance, weigh out 0.1000-0.1500g of your complex salt and place in a 250 mL Erlenmeyer flask.
2. Dissolve the sample in 10 mL of 6 M H₂SO₄. You may need to warm this solution to dissolve your sample.
3. Dilute the sample to approximately 75 mL with distilled water. Heat this solution to 80-90 °C and titrate as you did in part A. The endpoint will now have a peach colour instead of a pink colour and this is because of the iron(III) ions which when in solution give a yellow colour.
4. Repeat this procedure two more times.
5. Use the value for the blank that you obtained from Part IIA.

Part C: Determination of the amount of Iron in the Complex Salt

Since the iron in the complex salt is iron(III), the iron will be reduced to iron(II) and then oxidized with KMnO₄ to iron(III). The titration reaction is



Procedure

1. Using the **analytical** balance, weigh out 0.6000-0.7000g of your complex salt and place in a 250 mL beaker.
2. Add 25 mL of distilled water and 10 mL of 6M H₂SO₄. Heat this solution carefully to aid in dissolving your complex salt.
3. To get rid of the oxalate ions you must heat the solution to near boiling and add 3% KMnO₄ (NOT YOUR STANDARDIZED SOLUTION) with a medicine dropper until the solution remains purple or brown. This may require several millilitres but there is no need to be careful about measuring this amount. Of course you may want to make a note of how many drops it did take to save time for the next two titrations. Be sure not to add too much and make sure any droplets of 3% KMnO₄ on the walls of the flask are swirled into the solution.
4. Heat this solution to near boiling, take the beaker off the heat and place it under the hood. CAREFULLY add 4g of mossy zinc and stir with a stirring rod to avoid splashing. The solution will turn from a brown solution to a yellow solution to a colourless solution. You want all of the yellow colour to disappear but you do not want to add a large excess of zinc. The zinc is reducing the iron(III) in the complex salt to iron(II).

5. Cover the solution and wait to see if the yellow colour disappears. In the meantime, prepare to filter off the excess zinc (step 7). If the colour is not disappearing, add a little more zinc dust. When the solution is colourless, you have converted all of the yellow iron(III) to colourless iron(II).
6. Quickly filter the excess zinc because iron(II) tends to oxidize to iron(III) in the presence of air.
7. Set up a funnel and use regular qualitative filter paper. Set a clean 250 mL Erlenmeyer flask under the funnel. Hold the hot beaker with a paper towel and pour the hot solution through the filter paper. Using your wash bottle, rinse the beaker several times with distilled water onto the filter paper.
8. Reheat the Erlenmeyer flask to near boiling if necessary. In the meantime, fill the buret with your KMnO_4 solution and record the initial volume and begin titrating as soon as the solution is near boiling.
9. Start the titration adding KMnO_4 at a fast drip, swirling constantly. You do not want the solution to become cloudy. Slow down the addition of KMnO_4 , but keep swirling at all times, when you are approaching the endpoint. The endpoint is determined by a colour change from clear yellow to a clear peach colour. This colour change should stay for 20 to 30 seconds.
10. Repeat this procedure two more times.
11. Use the value for the blank that you obtained in Part IIA.

Once you have determined the percentages of oxalate and iron in your complex salt, you can determine the empirical formula of your salt and then the theoretical and percent yield of your coordination compound.

The following is a sample calculation for empirical formula and theoretical and percent yield.

Suppose you determine the following mass percentages: Mass % Oxalate = 55.20%
 Mass % Iron = 6.900%

Assume 100 g of complex salt.

Convert each mass to moles:

$$\text{moles of oxalate} = 55.20 \text{ g of oxalate} \times \frac{1 \text{ mole of oxalate}}{88.02 \text{ g of oxalate}} = 0.6271 \text{ moles of oxalate}$$

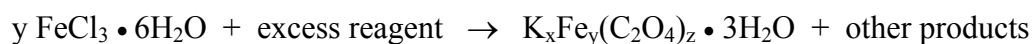
$$\text{moles of iron} = 6.900 \text{ g of iron} \times \frac{1 \text{ mole of iron}}{55.85 \text{ g of iron}} = 0.1235 \text{ moles of iron}$$

Divide each mole value by the smallest number of moles (0.1235).

$$\frac{0.6271}{0.1235} = 5.078$$

$$\frac{0.1235}{0.1235} = 1.000$$

There is a ratio of 5 moles of oxalate to one mole of iron. To write the empirical formula of $K_xFe_y(C_2O_4)_z \cdot 3H_2O$ you must know the charges of all the ions in the coordination compound. You have Fe^{3+} , $C_2O_4^{2-}$ and K^+ . So far you know $y = 1$ and that $z = 5$ that would mean x would have to equal 7 to give you the neutral compound. So your empirical formula is $K_7Fe(C_2O_4)_5 \cdot 3H_2O$. You are to assume that the empirical formula is also the molecular formula of the coordination compound. You can now calculate the theoretical and percent yields of your compound using the following equation.



You are to assume that $FeCl_3 \cdot 6H_2O$ is the limiting reagent. Balance the iron on either side of this equation and you get



The mole ratio between $FeCl_3 \cdot 6H_2O$ and $K_7Fe(C_2O_4)_5 \cdot 3H_2O$ is 1:1 so
moles of $FeCl_3 \cdot 6H_2O$ = moles of $K_7Fe(C_2O_4)_5 \cdot 3H_2O$

$$\text{moles of } FeCl_3 \cdot 6H_2O = \text{mass of } FeCl_3 \cdot 6H_2O / \text{MW of } FeCl_3 \cdot 6H_2O$$

$$\text{Theoretical yield} = \text{moles of } K_7Fe(C_2O_4)_5 \cdot 3H_2O \times \text{MW of } K_7Fe(C_2O_4)_5 \cdot 3H_2O$$

$$\% \text{ Yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

Results

Part I Synthesis of the Complex Salt containing Iron

Mass of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ _____

Mass of complex salt _____

Part II Analyzing the Complex Salt

Part A Standardizing a Potassium Permanganate Solution

Write the titration reaction performed in Part A in the space below.

Titration	1	2	3
Mass of $\text{Na}_2\text{C}_2\text{O}_4$ in sample	_____	_____	_____
Final Buret Reading	_____	_____	_____
Initial Buret Reading	_____	_____	_____
Volume of KMnO_4 used	_____	_____	_____
Volume of KMnO_4 used for blank		_____	
Corrected volume of KMnO_4 used	_____	_____	_____
Moles of $\text{Na}_2\text{C}_2\text{O}_4$	_____	_____	_____
Moles of $\text{C}_2\text{O}_4^{2-}$	_____	_____	_____
Moles of MnO_4^- used (look at titration reaction above)	_____	_____	_____
Moles of KMnO_4 used	_____	_____	_____
Molarity of KMnO_4	_____	_____	_____
Average Molarity of KMnO_4		_____	

Sample calculations of just ONE trial are to be shown in your laboratory notebook.

Part B Determination of the amount of Oxalate in the Complex Salt

Write the titration reaction for Part B in the space below.

Titration	1	2	3
Mass of Complex Salt	_____	_____	_____
Final Buret Reading	_____	_____	_____
Initial Buret Reading	_____	_____	_____
Volume of KMnO_4 used	_____	_____	_____
Volume of KMnO_4 used for blank		_____	
Corrected volume of KMnO_4 used	_____	_____	_____
Average Molarity of KMnO_4		_____	
Moles of MnO_4^- used	_____	_____	_____
Moles of $\text{C}_2\text{O}_4^{2-}$ in sample (use reaction above)	_____	_____	_____
Mass of $\text{C}_2\text{O}_4^{2-}$ in sample	_____	_____	_____
Mass % of $\text{C}_2\text{O}_4^{2-}$	_____	_____	_____
Average Mass % $\text{C}_2\text{O}_4^{2-}$		_____	

Sample calculations of just ONE trial are to be shown in your laboratory notebook.

Part C: Determination of the amount of Iron in the Complex Salt

Write the titration reaction for Part C in the space below.

Titration	1	2	3
Mass of Complex Salt	_____	_____	_____
Final Buret Reading	_____	_____	_____
Initial Buret Reading	_____	_____	_____
Volume of KMnO_4 used	_____	_____	_____
Volume of KMnO_4 used for blank		_____	
Corrected volume of KMnO_4 used	_____	_____	_____
Average Molarity of KMnO_4		_____	
Moles of MnO_4^- used	_____	_____	_____
Moles of Fe^{3+} in sample (use reaction above)	_____	_____	_____
Mass of Fe^{3+} in sample	_____	_____	_____
Mass % of Fe^{3+}	_____	_____	_____
Average Mass % Fe^{3+}	_____		

Sample calculations of just ONE trial are to be shown in your laboratory notebook.

Empirical Formula of Coordination Compound _____

Theoretical Yield of Coordination Compound _____

% Yield of Coordination Compound _____

Sample calculations are to be shown in your laboratory notebook.